

Likelihood & Maximum Likelihood

INFO 521 · Module 3 · Lecture A — From Loss to Likelihood

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Adapted with permission from lecture materials by Clayton Morrison, University of Arizona.

Learning outcomes

By the end of Lecture A you should be able to:

1. **State the linear-Gaussian noise model** and read it generatively.
2. **Write the likelihood** $p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \sigma^2)$ and the log-likelihood $\ell(\mathbf{w})$.
3. **Derive the MLE** — and show $\hat{\mathbf{w}}_{\text{MLE}}$ **is** the least-squares $\hat{\mathbf{w}}$ from Module 1.
4. **Estimate the noise** — $\hat{\sigma}^2$ — and explain why it is **biased**.

Why squared error? (the debt from Module 1)

Module 1 *chose* the mean-squared-error loss

$$\mathcal{L}(\mathbf{w}) = \frac{1}{N} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2$$

because it was smooth and penalized big misses. Reasonable — but a **choice**.

Today we *earn* it: squared error is what a **Gaussian noise model** implies. The accent hint on m1a's loss slide was this lecture.

The noise model

Assume each patient's systolic BP is the linear signal **plus Gaussian noise**:

$$y_n = \mathbf{w}^\top \mathbf{x}_n + \varepsilon_n, \quad \varepsilon_n \sim \mathcal{N}(0, \sigma^2) \text{ independently.}$$

Equivalently, each observation is a **draw from a Gaussian centered on the line**:

$$p(y_n \mid \mathbf{x}_n, \mathbf{w}, \sigma^2) = \mathcal{N}(y_n \mid \mathbf{w}^\top \mathbf{x}_n, \sigma^2).$$

This is a **generative story**: fix $\mathbf{w}, \sigma^2$, and the model can *produce* datasets — the picture m3b's repeated-datasets schematic draws.

The Gaussian, in one slide

$$\mathcal{N}(y \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y - \mu)^2}{2\sigma^2}\right)$$

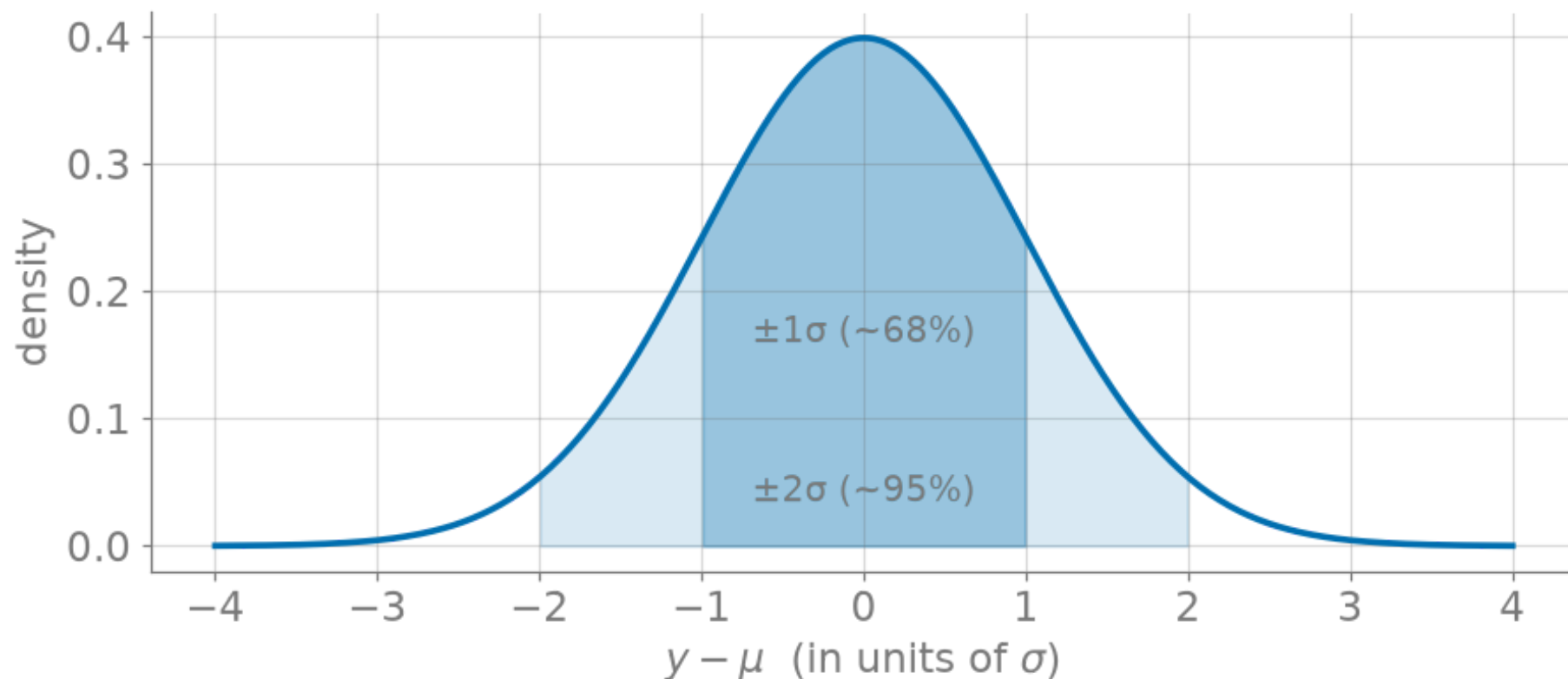


Figure 1: The Gaussian (illustrative): mean μ sets the center, σ sets the spread — ~68% within $\pm 1\sigma$, ~95% within $\pm 2\sigma$.

The likelihood

The patients are independent, so the probability of the **whole dataset** is a product — read as a **function of the parameters** with the data fixed:

$$p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \sigma^2) = \prod_{n=1}^N \mathcal{N}(y_n \mid \mathbf{w}^\top \mathbf{x}_n, \sigma^2)$$

- Data $(\mathbf{X}, \mathbf{y})$: **fixed** — it already happened.
- Parameters $(\mathbf{w}, \sigma^2)$: **free** — the knobs we turn.
- **The likelihood scores the knobs**: parameters under which the observed data are *more probable* score higher.

Maximum likelihood: pick the parameters that make the data you actually saw as probable as possible.

The log-likelihood

Products of exponentials beg for a logarithm — we write the log-likelihood $\ell(\mathbf{w}) = \log p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}, \sigma^2)$, reserving $\mathcal{L}$ for losses:

$$\ell(\mathbf{w}) = -\frac{N}{2} \log 2\pi - N \log \sigma - \frac{1}{2\sigma^2} \sum_{n=1}^N (y_n - \mathbf{w}^\top \mathbf{x}_n)^2$$

Look at the $\mathbf{w}$ -dependent part: it is **—(sum of squared errors), scaled**. Maximizing ℓ in $\mathbf{w}$ is minimizing squared error:

$$\arg \max_{\mathbf{w}} \ell(\mathbf{w}) = \arg \min_{\mathbf{w}} \mathcal{L}(\mathbf{w}).$$

The likelihood surface over NHANES

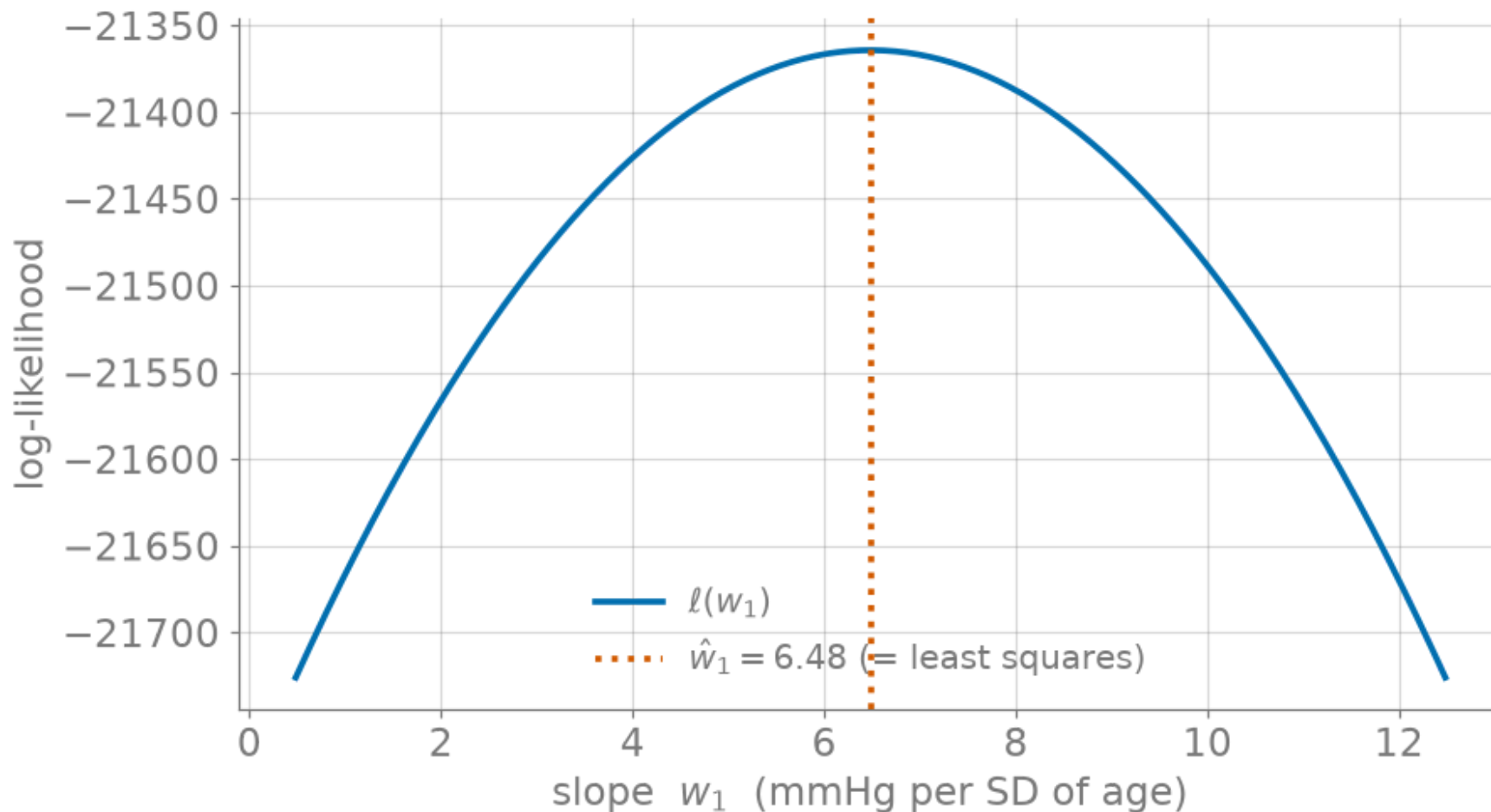


Figure 2: $\ell(w_1)$ for SBP on standardized age (intercept at $\hat{w}_0$): the peak IS the Module-1 least-squares slope.

MLE for $\mathbf{w}$: an old friend

Set the gradient of ℓ in $\mathbf{w}$ to zero:

$$\frac{\partial \ell}{\partial \mathbf{w}} = \frac{1}{\sigma^2} (\mathbf{X}^\top \mathbf{y} - \mathbf{X}^\top \mathbf{X} \mathbf{w}) = \mathbf{0} \quad \implies \quad \boxed{\hat{\mathbf{w}}_{\text{MLE}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}}$$

The **normal equations again** — σ^2 cancels entirely.

Least squares = maximum likelihood under Gaussian noise. Same $\hat{\mathbf{w}}$, better foundation: now it comes from a model, not a preference.

Dividend 1: the noise gets estimated too

The loss treated the scatter around the line as a nuisance. The likelihood makes it a **parameter**.
Setting $\partial\ell/\partial\sigma = 0$:

$$\hat{\sigma}^2 = \frac{1}{N} \sum_{n=1}^N (y_n - \hat{\mathbf{w}}^\top \mathbf{x}_n)^2 = \frac{1}{N} (\mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{X} \hat{\mathbf{w}})$$

— the **average squared residual** at the fitted line.

For SBP on standardized age: $\hat{\sigma} \approx 15.9$ mmHg — the model's honest “give or take” for one patient.

Model check: are the residuals Gaussian-ish?

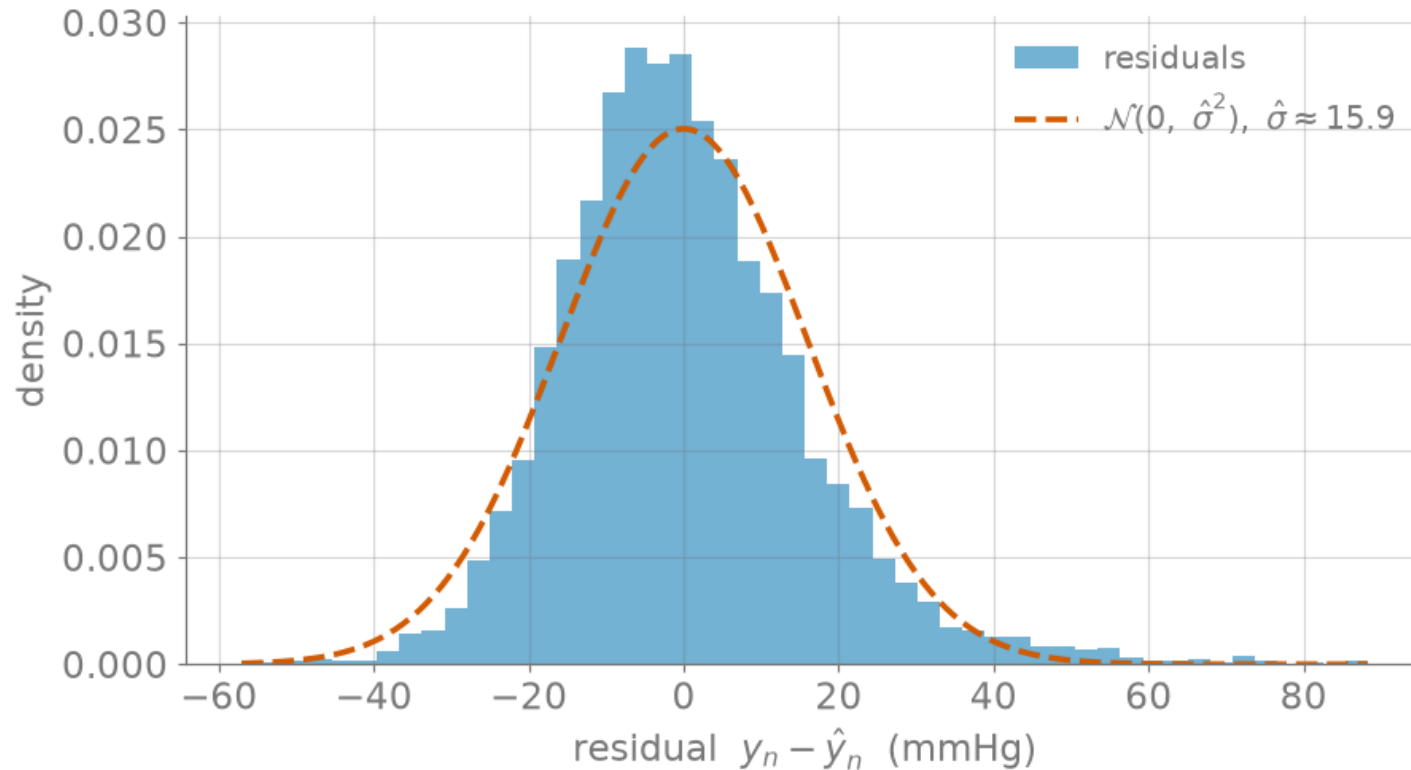


Figure 3: NHANES residuals vs. the fitted $\mathcal{N}(0, \hat{\sigma}^2)$: bell-ish, but right-skewed — models are assumptions, check them.

Dividend 2 — with a catch: $\hat{\sigma}^2$ is biased

Averaging over repeated datasets from the generating process:

$$\mathbb{E}[\hat{\sigma}^2] = \frac{N-D}{N} \sigma^2 < \sigma^2 \quad (D \text{ fitted parameters})$$

- The fit **chases the sample**: residuals from $\hat{\mathbf{w}}$ are *systematically smaller* than the true errors from $\mathbf{w}$.
- Each fitted parameter eats one *degree of freedom* — hence the $N-D$.
- The classic repair divides by $N-D$ instead of N ; at $N \approx 5,102$, $D = 2$, the difference is invisible — at $N = 10$ it is not.

Seeing the bias

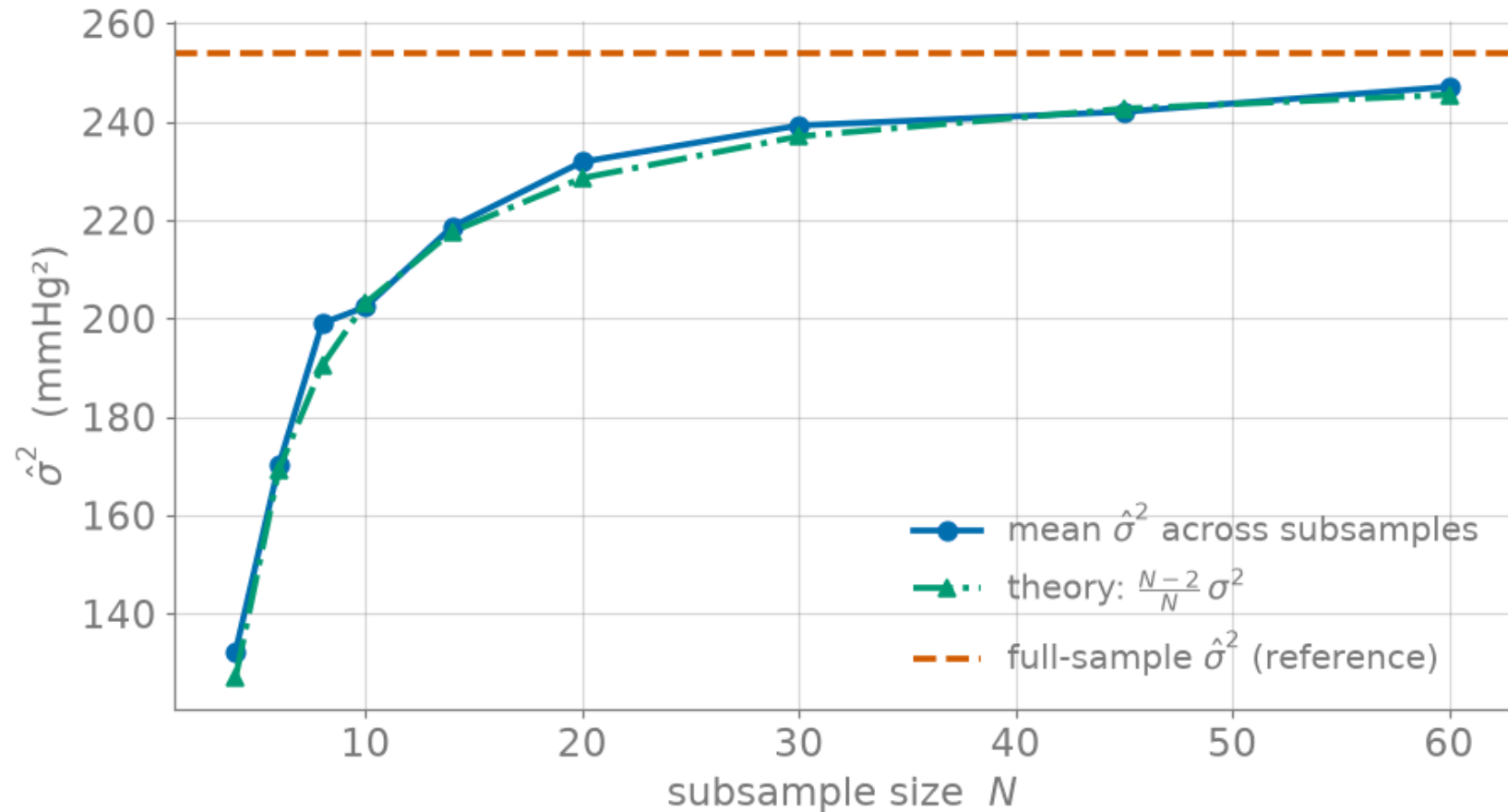


Figure 4: Mean $\hat{\sigma}^2$ over 2,000 NHANES subsamples per size: small samples systematically UNDER-estimate the noise, tracking $(N-2)/N \cdot \sigma^2$.

What maximum likelihood buys

One framework, three purchases:

1. **A principled loss.** Squared error stops being a taste and becomes a *consequence* (of Gaussian noise). Different noise model $\Rightarrow$ different loss.
2. **Noise as a first-class estimate.** $\hat{\sigma}^2$ quantifies patient-level scatter — the loss view had nowhere to put it.
3. **A road to uncertainty.** The *curvature* of ℓ at the peak says how sharply the data pin down $\hat{\mathbf{w}}$ — that is **Lecture B**.

And one caution: MLE estimators are not automatically unbiased ($\hat{\sigma}^2$ wasn't) — properties get **checked**, not assumed.

Recap & bridge

- **Noise model:** $y_n = \mathbf{w}^\top \mathbf{x}_n + \varepsilon_n$, $\varepsilon_n \sim \mathcal{N}(0, \sigma^2)$.
- **Likelihood** $\rightarrow \ell(\mathbf{w})$: maximizing it **is** minimizing $\mathcal{L}$ — least squares, re-earned.
- $\hat{\sigma}^2 = \frac{1}{N} \sum_n r_n^2$ — biased by $(N-D)/N$; repair with $N-D$.

Next (Lecture B): how *uncertain* is $\hat{\mathbf{w}}$? The Hessian of ℓ — the same $\mathbf{X}^\top \mathbf{X}$ yet again — becomes the Fisher information, and uncertainty gets a formula.